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A Demo of Interactive Thematic Data Collection on the Live Web

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> Many web agents stop at an answer. *Collection building continues for total coverage.*

01 Building Thematic Collections

Real workflows need **thematic web-data collection**: gathering large numbers of topically relevant documents scattered across the live web — municipal policy pages, university programs, job postings — rather than synthesizing one answer.

Success depends on **exploration efficiency and coverage across disjoint regions of the web graph**. Classical focused crawlers have the right objective but rely on fixed seeds, static relevance estimators, heuristic traversal and give the user no way to steer discovery.

02 Why not a deep-research agent?

ANSWER-ORIENTED AGENT	THEMATIC COLLECTION SYSTEM
Gathers evidence sufficient for one response or report	**Builds a comprehensive set of relevant documents**
Optimized for early stopping	**Long-horizon, budget-driven exploration**
Explores locally around the query	**Re-enters disjoint regions of the web graph**
User writes a prompt, reads a report	**User stays in the loop: corrects labels, adds guidance**
Returns a synthesized answer	**Returns a thematic collection with provenance artifacts**

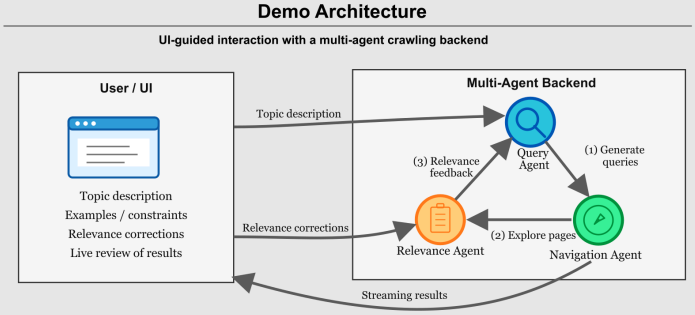
03 The dynamic re-seeding loop

The user starts from a natural-language theme description plus optional example URLs; the loop then alternates **local traversal** with **search-driven re-entry** through freshly generated queries.

ALGORITHM 1 · DYNAMIC RE-SEEDING LOOP

```
require theme t, examples E, crawl budget B
frontier F ← ∅,  visited V ← ∅,  relevant R ← ∅
while B not exhausted do
    Q ← QueryAgent(t, E, R)
    C ← Search(Q)                ▷ candidate entry points
    F ← F ∪ {c ∈ C | c ∉ V}
    while F ≠ ∅ and local budget remains do
        p ← highest-priority page in F
        visit p;
        V ← V ∪ {p}
        (p, s, r) ← RelevanceAgent(p, t, E, R)
        record page, score, rationale in R
        if user feedback available then
            update R with corrections and guidance
        L ← ExtractLinks(p)
        L' ← NavigationAgent(L, p, t, R)
        F ← F ∪ {ℓ ∈ L' | ℓ ∉ V}
```

04 Architecture



A FastAPI async backend handles sessions, steering, and artifact export; the interface is React. All three agents run on the DeepSeek API and exchange strict JSON.

Query Agent	Generates search queries that that identify promising entry points based on the user-provided collection theme, previously issued queries, and the results of previous queries
Relevance Agent	Evaluates whether visited pages are relevant to the collection theme based on page title, extracted content, collection theme, example page content, and user-provided feedback from previous pages
Navigation Agent	Ranks candidate links likelihood of leading to additional relevant pages given the surrounding page content and previously selected links

05 Demo scenario: the faculty job hunt

Postings for one field are spread over dozens of job boards and department sites, with no guarantee any single board carries a given job.

1. User enters the theme: `Faculty positions in Computer Science at US universities. Prioritize tenure-track.`
2. Optionally pin the search to particular boards or universities with the **Allowed Domains** filter.
3. The Query Agent issues its first searches. With **Review Before Crawl** on, the run pauses on the first batch.
4. Every hit is tenure-track only — too narrow. Leave the labels alone and write guidance instead: *"Broaden to all full-time faculty positions, including non-tenure-track and visiting."*
5. Both the Query and Navigation agents absorb the note; new queries and a re-ranked frontier widen the collection while you watch.
6. Pause, export the CSV — then resume the session later, or re-run it periodically to keep the collection current.

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CODE & ARTIFACTS
github.com/JustusWest/Web-Data-Discovery